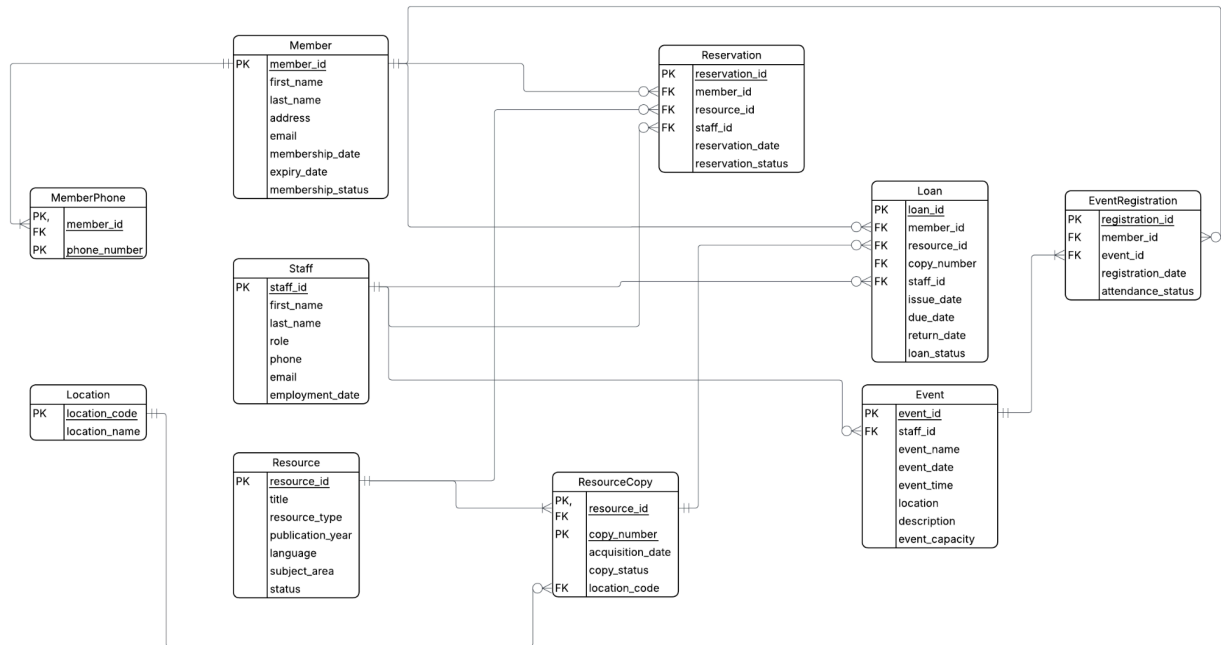


Library Database System Implementation

Relational Model / ERD from Part 1



1. Database Implementation in MySQL

1.1 Create Database

1.2 Create Tables (DDL)

1.3 Constraints and Relationships

```
CREATE DATABASE IF NOT EXISTS slv_library_db;
USE slv_library_db;

CREATE TABLE IF NOT EXISTS Member (
  member_id INT NOT NULL,
  first_name VARCHAR(50) NOT NULL,
  last_name VARCHAR(50) NOT NULL,
  address VARCHAR(255) NOT NULL,
  email VARCHAR(120) NOT NULL,
  membership_date DATE NOT NULL,
  expiry_date DATE NOT NULL,
  membership_status VARCHAR(20) NOT NULL DEFAULT 'Active',
  PRIMARY KEY (member_id),
  UNIQUE (email),
```

```

    CHECK (expiry_date >= membership_date),
    CHECK (membership_status IN ('Active', 'Expired', 'Suspended'))
);

CREATE TABLE IF NOT EXISTS Staff (
    staff_id INT NOT NULL,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    role VARCHAR(50) NOT NULL,
    phone VARCHAR(20) NOT NULL,
    email VARCHAR(120) NOT NULL,
    employment_date DATE NOT NULL,
    PRIMARY KEY (staff_id),
    UNIQUE (email)
);

CREATE TABLE IF NOT EXISTS Resource (
    resource_id INT NOT NULL,
    title VARCHAR(255) NOT NULL,
    resource_type VARCHAR(30) NOT NULL,
    publication_year YEAR,
    language VARCHAR(50) NOT NULL,
    subject_area VARCHAR(100) NOT NULL,
    status VARCHAR(20) NOT NULL DEFAULT 'Available',
    PRIMARY KEY (resource_id),
    CHECK (resource_type IN ('Book', 'Journal', 'Archive', 'eBook', 'Database')),
    CHECK (status IN ('Available', 'Unavailable', 'Archived'))
);

CREATE TABLE IF NOT EXISTS Location (
    location_code VARCHAR(20) NOT NULL,
    location_name VARCHAR(100) NOT NULL,
    PRIMARY KEY (location_code),
    UNIQUE (location_name)
);

CREATE TABLE IF NOT EXISTS ResourceCopy (
    resource_id INT NOT NULL,
    copy_number INT NOT NULL,
    acquisition_date DATE NOT NULL,
    copy_status VARCHAR(20) NOT NULL DEFAULT 'Available',
    location_code VARCHAR(20) NOT NULL,

```

```

PRIMARY KEY (resource_id, copy_number),
FOREIGN KEY (resource_id) REFERENCES Resource(resource_id),
FOREIGN KEY (location_code) REFERENCES Location(location_code),
CHECK (copy_number > 0),
CHECK (copy_status IN ('Available', 'On Loan', 'Reserved', 'Maintenance'))
);

CREATE TABLE IF NOT EXISTS MemberPhone (
    member_id INT NOT NULL,
    phone_number VARCHAR(20) NOT NULL,
    PRIMARY KEY (member_id, phone_number),
    FOREIGN KEY (member_id) REFERENCES Member(member_id)
);

CREATE TABLE IF NOT EXISTS Loan (
    loan_id INT NOT NULL,
    member_id INT NOT NULL,
    resource_id INT NOT NULL,
    copy_number INT NOT NULL,
    staff_id INT NOT NULL,
    issue_date DATE NOT NULL,
    due_date DATE NOT NULL,
    return_date DATE NULL,
    loan_status VARCHAR(20) NOT NULL DEFAULT 'Issued',
    PRIMARY KEY (loan_id),
    FOREIGN KEY (member_id) REFERENCES Member(member_id),
    FOREIGN KEY (staff_id) REFERENCES Staff(staff_id),
    FOREIGN KEY (resource_id, copy_number) REFERENCES ResourceCopy(resource_id, copy_number),
    CHECK (due_date >= issue_date),
    CHECK (return_date IS NULL OR return_date >= issue_date),
    CHECK (loan_status IN ('Issued', 'Returned', 'Overdue'))
);

CREATE TABLE IF NOT EXISTS Reservation (
    reservation_id INT NOT NULL,
    member_id INT NOT NULL,
    resource_id INT NOT NULL,
    staff_id INT NOT NULL,
    reservation_date DATE NOT NULL,
    reservation_status VARCHAR(20) NOT NULL DEFAULT 'Pending',
    PRIMARY KEY (reservation_id),
    FOREIGN KEY (member_id) REFERENCES Member(member_id),

```

```

FOREIGN KEY (resource_id) REFERENCES Resource(resource_id),
FOREIGN KEY (staff_id) REFERENCES Staff(staff_id),
CHECK (reservation_status IN ('Pending', 'Confirmed', 'Cancelled', 'Completed'))
);

CREATE TABLE IF NOT EXISTS `Event` (
  event_id INT NOT NULL,
  staff_id INT NOT NULL,
  event_name VARCHAR(150) NOT NULL,
  event_date DATE NOT NULL,
  event_time TIME NOT NULL,
  location VARCHAR(120) NOT NULL,
  description VARCHAR(500),
  event_capacity INT NOT NULL,
  PRIMARY KEY (event_id),
  FOREIGN KEY (staff_id) REFERENCES Staff(staff_id),
  CHECK (event_capacity > 0)
);

CREATE TABLE IF NOT EXISTS EventRegistration (
  registration_id INT NOT NULL,
  member_id INT NOT NULL,
  event_id INT NOT NULL,
  registration_date DATE NOT NULL,
  attendance_status VARCHAR(20) NOT NULL DEFAULT 'Registered',
  PRIMARY KEY (registration_id),
  FOREIGN KEY (member_id) REFERENCES Member(member_id),
  FOREIGN KEY (event_id) REFERENCES `Event`(event_id),
  CHECK (attendance_status IN ('Registered', 'Attended', 'No Show', 'Cancelled'))
);

```

2. Data Population

2.1 Insert Records into Each Table

```

USE slv_library_db;

INSERT INTO Member
(member_id, first_name, last_name, address, email, membership_date, expiry_date, membership_status)
VALUES
(101, 'Amelia', 'Brown', '12 Collins St, Melbourne', 'amelia.brown@email.com', '2025-01-15', '2026-01-15', 'Active'),

```

```
(102, 'Noah', 'Wilson', '45 Swanston St, Melbourne', 'noah.wilson@email.com', '2024-11-01', '2025-11-01', 'Expired'),
(103, 'Olivia', 'Taylor', '8 Lygon St, Carlton', 'olivia.taylor@email.com', '2025-06-10', '2026-06-10', 'Active'),
(104, 'Liam', 'Johnson', '77 Chapel St, Prahran', 'liam.johnson@email.com', '2025-03-05', '2026-03-05', 'Suspended'),
(105, 'Charlotte', 'Davis', '21 High St, Northcote', 'charlotte.davis@email.com', '2025-09-20', '2026-09-20', 'Active');
```

INSERT INTO Staff

(staff_id, first_name, last_name, role, phone, email, employment_date)

VALUES

```
(201, 'Ethan', 'Miller', 'Librarian', '03-9000-1001', 'ethan.miller@slv.vic.gov.au', '2020-02-10'),
(202, 'Sophia', 'Anderson', 'Archivist', '03-9000-1002', 'sophia.anderson@slv.vic.gov.au', '2019-07-01'),
(203, 'James', 'Thomas', 'Events Coordinator', '03-9000-1003', 'james.thomas@slv.vic.gov.au', '2021-01-18'),
(204, 'Mia', 'Jackson', 'Membership Officer', '03-9000-1004', 'mia.jackson@slv.vic.gov.au', '2022-05-22'),
(205, 'Lucas', 'White', 'Digital Services Officer', '03-9000-1005', 'lucas.white@slv.vic.gov.au', '2023-03-14');
```

INSERT INTO Resource

(resource_id, title, resource_type, publication_year, language, subject_area, status)

VALUES

```
(301, 'Australian History Essentials', 'Book', 2020, 'English', 'History', 'Available'),
(302, 'Modern Data Systems', 'Journal', 2023, 'English', 'Technology', 'Available'),
(303, 'Victoria Archives Collection', 'Archive', 2018, 'English', 'Government Records', 'Archived'),
(304, 'Digital Research Methods', 'eBook', 2024, 'English', 'Research', 'Available'),
(305, 'Academic Search Premium', 'Database', 2022, 'English', 'Multi-disciplinary', 'Unavailable');
```

INSERT INTO Location

(location_code, location_name)

VALUES

```
('L01', 'Reading Room A'),
('L02', 'Archive Storage 1'),
('L03', 'Digital Access Zone'),
('L04', 'Reference Section'),
('L05', 'Exhibition Hall');
```

INSERT INTO ResourceCopy

(resource_id, copy_number, acquisition_date, copy_status, location_code)

VALUES

```
(301, 1, '2021-02-15', 'Available', 'L01'),
(302, 1, '2023-08-10', 'On Loan', 'L04'),
(303, 1, '2019-01-05', 'Maintenance', 'L02'),
(304, 1, '2024-03-20', 'Reserved', 'L03'),
(305, 1, '2022-11-11', 'Available', 'L03');
```

INSERT INTO MemberPhone

```
(member_id, phone_number)
```

```
VALUES
```

```
(101, '+61411111111'),  
(102, '+61422222222'),  
(103, '+61433333333'),  
(104, '+61444444444'),  
(105, '+61455555555');
```

```
INSERT INTO Loan
```

```
(loan_id, member_id, resource_id, copy_number, staff_id, issue_date, due_date, return_date, loan_status)
```

```
VALUES
```

```
(401, 101, 301, 1, 201, '2026-03-01', '2026-03-15', '2026-03-12', 'Returned'),  
(402, 102, 302, 1, 202, '2026-03-20', '2026-04-03', NULL, 'Issued'),  
(403, 103, 303, 1, 203, '2026-02-01', '2026-02-14', '2026-02-20', 'Overdue'),  
(404, 104, 304, 1, 204, '2026-01-10', '2026-01-24', '2026-01-23', 'Returned'),  
(405, 105, 305, 1, 205, '2026-04-01', '2026-04-15', NULL, 'Issued');
```

```
INSERT INTO Reservation
```

```
(reservation_id, member_id, resource_id, staff_id, reservation_date, reservation_status)
```

```
VALUES
```

```
(501, 101, 305, 201, '2026-04-05', 'Pending'),  
(502, 102, 301, 202, '2026-04-02', 'Confirmed'),  
(503, 103, 304, 203, '2026-03-30', 'Cancelled'),  
(504, 104, 302, 204, '2026-04-01', 'Completed'),  
(505, 105, 303, 205, '2026-04-06', 'Pending');
```

```
INSERT INTO `Event`
```

```
(event_id, staff_id, event_name, event_date, event_time, location, description, event_capacity)
```

```
VALUES
```

```
(601, 203, 'Family History Workshop', '2026-05-12', '10:00:00', 'Theatrette', 'Workshop on genealogy research tools',  
40),  
(602, 201, 'Rare Books Exhibition Tour', '2026-05-18', '14:00:00', 'Exhibition Hall', 'Guided tour of rare books  
collection', 30),  
(603, 202, 'Archive Preservation Talk', '2026-06-02', '11:30:00', 'Seminar Room 1', 'Talk on archival preservation  
practices', 50),  
(604, 204, 'New Member Orientation', '2026-04-20', '09:30:00', 'Reading Room A', 'Orientation for new members', 60),  
(605, 205, 'Digital Databases Training', '2026-05-25', '15:00:00', 'Digital Access Zone', 'Hands-on database search  
training', 35);
```

```
INSERT INTO EventRegistration
```

```
(registration_id, member_id, event_id, registration_date, attendance_status)
```

```
VALUES
```


92
93 SELECT * FROM Member;
94 SELECT * FROM Staff;
95 SELECT * FROM Resource;
96 SELECT * FROM Location;
SELECT * FROM ResourceCopy;
SELECT * FROM MemberPhone;
99 SELECT * FROM Loan;
SELECT * FROM Reservation;
SELECT * FROM 'Event';
SELECT * FROM EventRegistration;

100% 24:95

Result Grid Filter Rows: Search Edit: Export/Import:

resource_id	title	resource_ty...	publication_ye...	language	subject_area	status
301	Australian History Essentials	Book	2020	English	History	Available
302	Modern Data Systems	Journal	2023	English	Technology	Available
303	Victoria Archives Collection	Archive	2018	English	Government Records	Archived
304	Digital Research Methods	eBook	2024	English	Research	Available
305	Academic Search Premium	Database	2022	English	Multi-disciplinary	Unavailable
HULL	HULL	HULL	HULL	HULL	HULL	HULL

92
93 SELECT * FROM Member;
94 SELECT * FROM Staff;
95 SELECT * FROM Resource;
96 SELECT * FROM Location;
97 SELECT * FROM ResourceCopy;
SELECT * FROM MemberPhone;
SELECT * FROM Loan;
100 SELECT * FROM Reservation;
SELECT * FROM 'Event';
SELECT * FROM EventRegistration;

100% 24:96

Result Grid Filter Rows: Search Edit: Export/Import:

location_co...	location_name
L02	Archive Storage 1
L03	Digital Access Zone
L05	Exhibition Hall
L01	Reading Room A
L04	Reference Section
HULL	HULL

92
93 SELECT * FROM Member;
94 SELECT * FROM Staff;
95 SELECT * FROM Resource;
96 SELECT * FROM Location;
97 SELECT * FROM ResourceCopy;
98 SELECT * FROM MemberPhone;
SELECT * FROM Loan;
SELECT * FROM Reservation;
SELECT * FROM 'Event';
SELECT * FROM EventRegistration;

100% 28:97

Result Grid Filter Rows: Search Edit: Export/Import:

resource_id	copy_number	acquisition_da...	copy_status	location_co...
301	1	2021-02-15	Available	L01
302	1	2023-08-10	On Loan	L04
303	1	2019-01-05	Maintenance	L02
304	1	2024-03-20	Reserved	L03
305	1	2022-11-11	Available	L03
HULL	HULL	HULL	HULL	HULL


```

92
93 SELECT * FROM Member;
94 SELECT * FROM Staff;
95 SELECT * FROM Resource;
96 SELECT * FROM Location;
97 SELECT * FROM ResourceCopy;
98 SELECT * FROM MemberPhone;
99 SELECT * FROM Loan;
    SELECT * FROM Reservation;
    SELECT * FROM 'Event';
    SELECT * FROM EventRegistration;

```

100% 27:98

Result Grid Filter Rows: Search Edit: Export/Import:

member_id	phone_number
101	+61411111111
102	+61422222222
103	+61433333333
104	+61444444444
105	+61455555555
NULL	NULL

```

92
93 SELECT * FROM Member;
94 SELECT * FROM Staff;
95 SELECT * FROM Resource;
96 SELECT * FROM Location;
97 SELECT * FROM ResourceCopy;
98 SELECT * FROM MemberPhone;
99 SELECT * FROM Loan;
100 SELECT * FROM Reservation;
    SELECT * FROM 'Event';
    SELECT * FROM EventRegistration;

```

100% 20:99

Result Grid Filter Rows: Search Edit: Export/Import:

loan_id	member_id	resource_id	copy_number	staff_id	issue_date	due_date	return_date	loan_status
401	101	301	1	201	2026-03-01	2026-03-15	2026-03-12	Returned
402	102	302	1	202	2026-03-20	2026-04-03	NULL	Issued
403	103	303	1	203	2026-02-01	2026-02-14	2026-02-20	Overdue
404	104	304	1	204	2026-01-10	2026-01-24	2026-01-23	Returned
405	105	305	1	205	2026-04-01	2026-04-15	NULL	Issued
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

```

92
93 SELECT * FROM Member;
94 SELECT * FROM Staff;
95 SELECT * FROM Resource;
96 SELECT * FROM Location;
97 SELECT * FROM ResourceCopy;
98 SELECT * FROM MemberPhone;
99 SELECT * FROM Loan;
100 SELECT * FROM Reservation;
101 SELECT * FROM 'Event';
    SELECT * FROM EventRegistration;

```

100% 27:100

Result Grid Filter Rows: Search Edit: Export/Import:

reservation_id	member_id	resource_id	staff_id	reservation_date	reservation_status
501	101	305	201	2026-04-05	Pending
502	102	301	202	2026-04-02	Confirmed
503	103	304	203	2026-03-30	Cancelled
504	104	302	204	2026-04-01	Completed
505	105	303	205	2026-04-06	Pending
NULL	NULL	NULL	NULL	NULL	NULL

```

92
93 SELECT * FROM Member;
94 SELECT * FROM Staff;
95 SELECT * FROM Resource;
96 SELECT * FROM Location;
97 SELECT * FROM ResourceCopy;
98 SELECT * FROM MemberPhone;
99 SELECT * FROM Loan;
100 SELECT * FROM Reservation;
101 SELECT * FROM 'Event';
102 SELECT * FROM EventRegistration;

```

100% 23:101

Result Grid Filter Rows: Search Edit: Export/Import:

event_id	staff_id	event_name	event_date	event_time	location	description	event_capaci...
601	203	Family History Workshop	2026-05-12	10:00:00	Theatrette	Workshop on genealogy research tools	40
602	201	Rare Books Exhibition Tour	2026-05-18	14:00:00	Exhibition Hall	Guided tour of rare books collection	30
603	202	Archive Preservation Talk	2026-06-02	11:30:00	Seminar Room 1	Talk on archival preservation practices	50
604	204	New Member Orientation	2026-04-20	09:30:00	Reading Room A	Orientation for new members	60
605	205	Digital Databases Training	2026-05-25	15:00:00	Digital Access Zone	Hands-on database search training	35
HULL	HULL	HULL	HULL	HULL	HULL	HULL	HULL

```

92
93 SELECT * FROM Member;
94 SELECT * FROM Staff;
95 SELECT * FROM Resource;
96 SELECT * FROM Location;
97 SELECT * FROM ResourceCopy;
98 SELECT * FROM MemberPhone;
99 SELECT * FROM Loan;
100 SELECT * FROM Reservation;
101 SELECT * FROM 'Event';
102 SELECT * FROM EventRegistration;
103

```

100% 33:102

Result Grid Filter Rows: Search Edit: Export/Import:

registration...	member_id	event_id	registration_d...	attendance_stat...
701	101	601	2026-04-10	Registered
702	102	602	2026-04-11	Cancelled
703	103	603	2026-04-12	Registered
704	104	604	2026-04-13	Attended
705	105	605	2026-04-14	No Show
HULL	HULL	HULL	HULL	HULL

3. Schema Modification

3.1 Add a New Column with Default Value

The table selected for modification is Reservation. The new column to be added is `priority_level`, which has a data type of `VARCHAR(20)` and a default value of 'Standard'.

```

USE slv_library_db;

ALTER TABLE Reservation
ADD COLUMN priority_level VARCHAR(20) NOT NULL DEFAULT 'Standard';

SELECT * FROM Reservation;

```

3.2 Verification of Default Values in Existing Rows

```

1 • USE slv_library_db;
2
3 • ALTER TABLE Reservation
4   ADD COLUMN priority_level VARCHAR(20) NOT NULL DEFAULT 'Standard';
5
6 • SELECT * FROM Reservation;

```

reservation_id	member_id	resource_id	staff_id	reservation_da...	reservation_stat...	priority_level
501	101	305	201	2026-04-05	Pending	Standard
502	102	301	202	2026-04-02	Confirmed	Standard
503	103	304	203	2026-03-30	Cancelled	Standard
504	104	302	204	2026-04-01	Completed	Standard
505	105	303	205	2026-04-06	Pending	Standard
HULL	HULL	HULL	HULL	HULL	HULL	HULL

4. Data Update Using Conditions

4.1 Update Newly Added Column for Selected Rows

The rows are filtered using the condition `reservation_status IN ('Pending', 'Confirmed')`, so only reservations with those statuses are updated. For those selected rows, the newly added column `priority_level` is set to 'High'.

```

USE slv_library_db;

UPDATE Reservation
SET priority_level = 'High'
WHERE reservation_id >= 0
AND reservation_status IN ('Pending', 'Confirmed');
SELECT * FROM Reservation;

```

4.2 Verification of Conditional Update Results

```

1 • USE slv_library_db;
2
3 • UPDATE Reservation
4   SET priority_level = 'High'
5   WHERE reservation_id >= 0
6     AND reservation_status IN ('Pending', 'Confirmed');
7
8 • SELECT * FROM Reservation;

```

reservation_id	member_id	resource_id	staff_id	reservation_da...	reservation_stat...	priority_level
501	101	305	201	2026-04-05	Pending	High
502	102	301	202	2026-04-02	Confirmed	High
503	103	304	203	2026-03-30	Cancelled	Standard
504	104	302	204	2026-04-01	Completed	Standard
505	105	303	205	2026-04-06	Pending	High
HULL	HULL	HULL	HULL	HULL	HULL	HULL

5. SQL Query Tasks

5.1 Query A: SELECT with INNER JOIN and ORDER BY

I want to list loan transactions with member names and resource titles, ordered by due date from earliest to latest.

```
SELECT
  l.loan_id,
  m.first_name,
  m.last_name,
  r.title,
  l.issue_date,
  l.due_date,
  l.loan_status
FROM Loan l
INNER JOIN Member m ON l.member_id = m.member_id
INNER JOIN Resource r ON l.resource_id = r.resource_id
ORDER BY l.due_date ASC;
```

```
1  -- 5(a) SELECT with INNER JOIN and ORDER BY
2  -- I want to list loan transactions with member names and resource titles, ordered by due date from earliest to latest.
3
4  • SELECT
5      l.loan_id,
6      m.first_name,
7      m.last_name,
8      r.title,
9      l.issue_date,
10     l.due_date,
11     l.loan_status
12 FROM Loan l
13 INNER JOIN Member m ON l.member_id = m.member_id
14 INNER JOIN Resource r ON l.resource_id = r.resource_id
15 ORDER BY l.due_date ASC;
```

100% 25:15

Result Grid

Filter Rows:

Search

Export:

	loan_id	first_name	last_name	title	issue_date	due_date	loan_status
	404	Liam	Johnson	Digital Research Methods	2026-01-10	2026-01-24	Returned
	403	Olivia	Taylor	Victoria Archives Collection	2026-02-01	2026-02-14	Overdue
	401	Amelia	Brown	Australian History Essentials	2026-03-01	2026-03-15	Returned
	402	Noah	Wilson	Modern Data Systems	2026-03-20	2026-04-03	Issued
	405	Charlotte	Davis	Academic Search Premium	2026-04-01	2026-04-15	Issued

5.2 Query B: SELECT with WHERE and IN

I want to retrieve reservations that are either pending or confirmed.

```
SELECT
  reservation_id,
  member_id,
  resource_id,
  reservation_date,
  reservation_status,
  priority_level
FROM Reservation
WHERE reservation_status IN ('Pending', 'Confirmed');
```

```

17 -- 5(b) SELECT with WHERE and IN
18 -- I want to retrieve reservations that are either pending or confirmed.
19
20 SELECT
21     reservation_id,
22     member_id,
23     resource_id,
24     reservation_date,
25     reservation_status,
26     priority_level
27 FROM Reservation
28 WHERE reservation_status IN ('Pending', 'Confirmed');
29

```

100% 54:28

Result Grid Filter Rows: Search Edit: Export/Import:

reservation_id	member_id	resource_id	reservation_date	reservation_status	priority_level
501	101	305	2026-04-05	Pending	High
502	102	301	2026-04-02	Confirmed	High
505	105	303	2026-04-06	Pending	High
NULL	NULL	NULL	NULL	NULL	NULL

5.3 Query C: SELECT Using a Date Function

I want to calculate the loan period in days for each loan.

```

SELECT
    loan_id,
    issue_date,
    due_date,
    DATEDIFF(due_date, issue_date) AS loan_period_days
FROM Loan;

```

```

30 -- 5(c) SELECT using a DATE function
31 -- I want to calculate the loan period in days for each loan.
32
33 SELECT
34     loan_id,
35     issue_date,
36     due_date,
37     DATEDIFF(due_date, issue_date) AS loan_period_days
38 FROM Loan;
39

```

100% 11:38

Result Grid Filter Rows: Search Export:

loan_id	issue_date	due_date	loan_period_days
401	2026-03-01	2026-03-15	14
402	2026-03-20	2026-04-03	14
403	2026-02-01	2026-02-14	13
404	2026-01-10	2026-01-24	14
405	2026-04-01	2026-04-15	14

5.4 Query D: Create and Test a VIEW with JOIN

I want to create a view that shows event registrations with member details and event details.

```

CREATE VIEW vw_event_registrations AS
SELECT
    er.registration_id,
    m.member_id,

```

```

CONCAT(m.first_name, ' ', m.last_name) AS member_name,
e.event_id,
e.event_name,
e.event_date,
er.attendance_status
FROM EventRegistration er
JOIN Member m ON er.member_id = m.member_id
JOIN `Event` e ON er.event_id = e.event_id;

SELECT * FROM vw_event_registrations;

```

```

40 -- 5(d) Create and test a VIEW with JOIN
41 -- I want to create a view that shows event registrations with member details and event details.
42
43 CREATE VIEW vw_event_registrations AS
44 SELECT
45     er.registration_id,
46     m.member_id,
47     CONCAT(m.first_name, ' ', m.last_name) AS member_name,
48     e.event_id,
49     e.event_name,
50     e.event_date,
51     er.attendance_status
52 FROM EventRegistration er
53 JOIN Member m ON er.member_id = m.member_id
54 JOIN `Event` e ON er.event_id = e.event_id;
55
56 SELECT * FROM vw_event_registrations;
57

```

100% 38:56

Result Grid Filter Rows: Search Export:

registration...	member_id	member_name	event_id	event_name	event_date	attendance_stat...
701	101	Amelia Brown	601	Family History Workshop	2026-05-12	Registered
702	102	Noah Wilson	602	Rare Books Exhibition Tour	2026-05-18	Cancelled
703	103	Olivia Taylor	603	Archive Preservation Talk	2026-06-02	Registered
704	104	Liam Johnson	604	New Member Orientation	2026-04-20	Attended
705	105	Charlotte Davis	605	Digital Databases Training	2026-05-25	No Show